



Applied Machine Learning

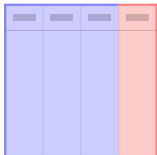
Machine Learning in R: MLR3 Resampling Strategies



Learning goals

- Resampling in MLR3
- Working with 'ResamplingResult' object

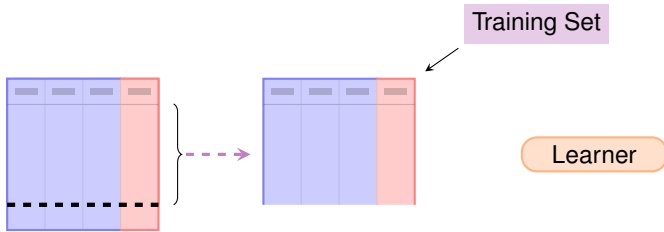
RESAMPLING



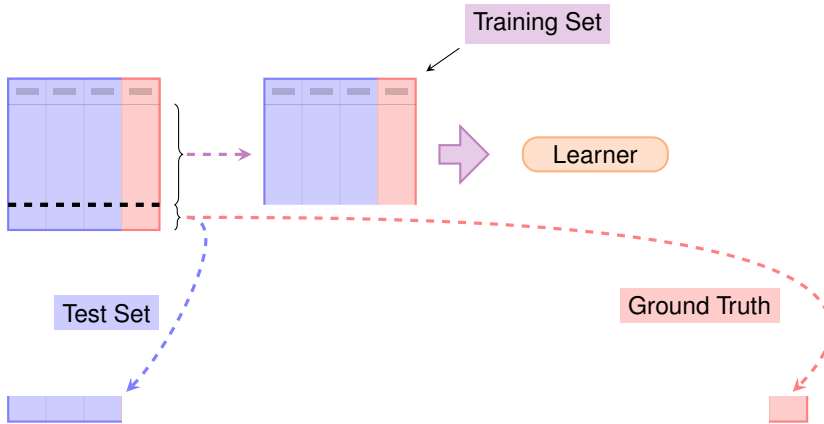
Learner



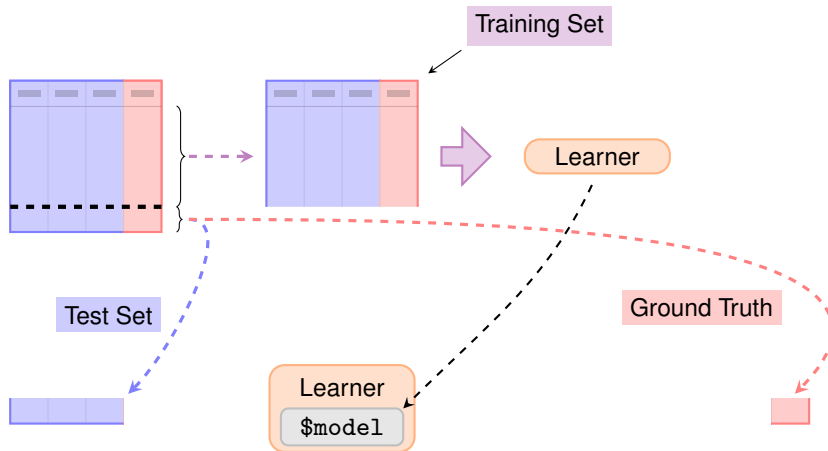
RESAMPLING



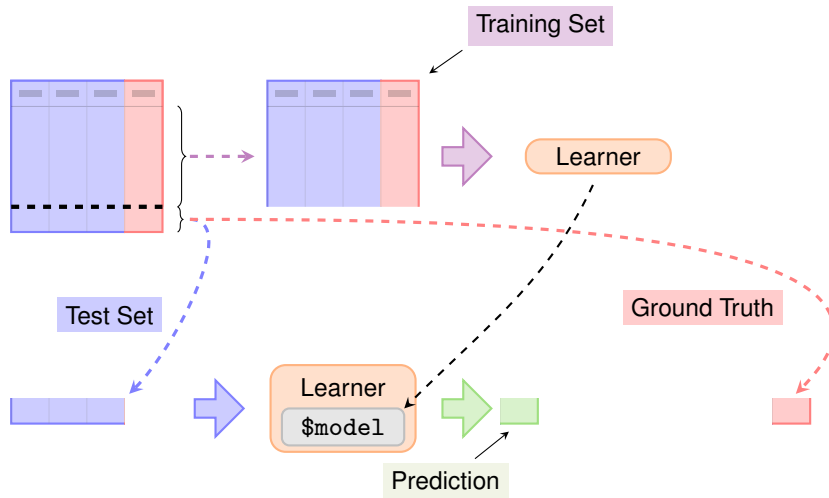
RESAMPLING

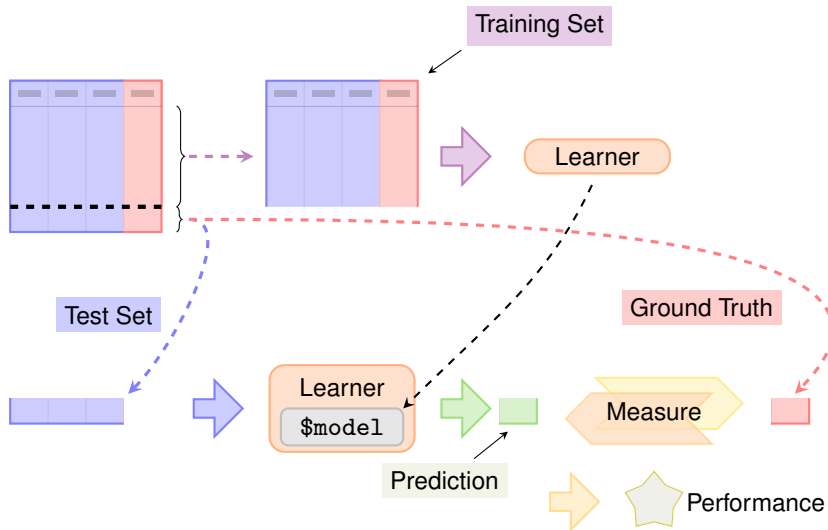


RESAMPLING

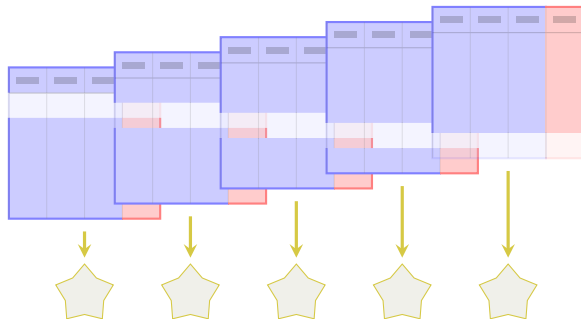


RESAMPLING

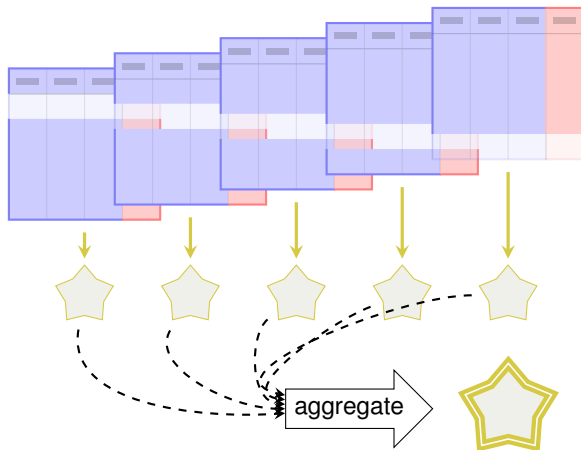




RESAMPLING



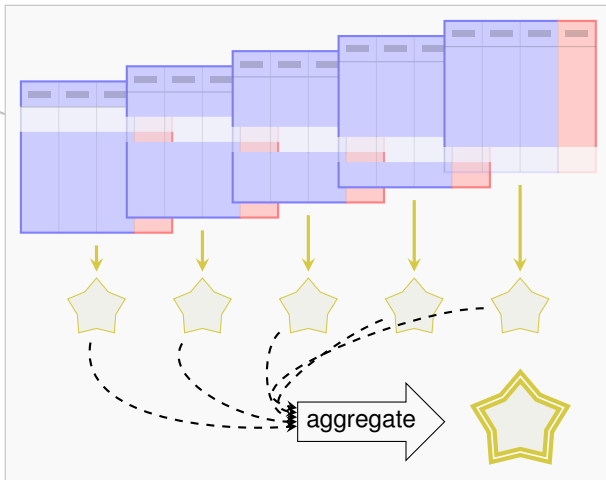
RESAMPLING



RESAMPLING



resample()



RESAMPLING

- Resample description: How to split the data

```
cv5 = rsmp("cv", folds = 5)
```



RESAMPLING



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```

- Use the `resample()` function for resampling:

```
task = as_task_classif(x = iris, target = "Species", id = "iris")  
learner = lrn("classif.rpart")  
rr = resample(task, learner, cv5)
```



- Resample description: How to split the data

```
cv5 = rsmp("cv", folds = 5)
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- Use the `resample()` function for resampling:

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task = as_task_classif(x = iris, target = "Species", id = "iris")
learner = lrn("classif.rpart")
rr = resample(task, learner, cv5)
```

- We get a `ResamplingResult` object:

```
print(rr)
#> <ResampleResult> with 5 resampling iterations
#>   task_id   learner_id resampling_id iteration warnings errors
#>   iris classif.rpart      cv          1          0          0
#>   iris classif.rpart      cv          2          0          0
#>   iris classif.rpart      cv          3          0          0
#>   iris classif.rpart      cv          4          0          0
#>   iris classif.rpart      cv          5          0          0
```

RESAMPLING RESULTS



What exactly is a ResamplingResult object?

Remember Prediction:

- Get a table representation using `as.data.table()`

```
rr_table = as.data.table(rr)

print(rr_table)

#           task           learner
#           <list>           <list>
# 1: <TaskClassif:iris> <LearnerClassifRpart:classif.rpart>
# 2: <TaskClassif:iris> <LearnerClassifRpart:classif.rpart>
# 3: <TaskClassif:iris> <LearnerClassifRpart:classif.rpart>
# 4: <TaskClassif:iris> <LearnerClassifRpart:classif.rpart>
# 5: <TaskClassif:iris> <LearnerClassifRpart:classif.rpart>
#   resampling iteration prediction
#   <list>      <int>      <list>
# 1: <ResamplingCV>      1 <PredictionClassif>
# 2: <ResamplingCV>      2 <PredictionClassif>
# 3: <ResamplingCV>      3 <PredictionClassif>
# 4: <ResamplingCV>      4 <PredictionClassif>
# 5: <ResamplingCV>      5 <PredictionClassif>
```

- Active bindings and functions that make information easily accessible

RESAMPLING RESULTS

- Calculate performance:

```
rr$aggregate(msr("classif.ce"))  
#> classif.ce  
#>          0.06
```



RESAMPLING RESULTS



- Calculate performance:

```
rr$aggregate(msr("classif.ce"))  
#> classif.ce  
#>          0.06
```

- Get predictions

```
rr$prediction()  
#> <PredictionClassif> for 150 observations:  
#>   row_ids   truth  response  
#>       2   setosa   setosa  
#>       7   setosa   setosa  
#>       9   setosa   setosa  
#> ---  
#>    141 virginica virginica  
#>    148 virginica virginica  
#>    149 virginica virginica
```


RESAMPLING

- Predictions of individual folds

```
predictions = rr$predictions()
predictions[[1]]

#> <PredictionClassif> for 30 observations:
#>      row_ids      truth  response
#>         2      setosa    setosa
#>         7      setosa    setosa
#>         9      setosa    setosa
#> ---
#>      135 virginica versicolor
#>      136 virginica  virginica
#>      142 virginica  virginica
```



RESAMPLING

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#> <PredictionClassif> for 30 observations:
#>      row_ids      truth  response
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#>          9      setosa    setosa
#> ---
#>      135 virginica versicolor
#>      136 virginica  virginica
#>      142 virginica  virginica
```

- Score of individual folds

```
scores = rr$score()
scores[1:3, c("iteration", "classif.ce")]

#>      iteration classif.ce
#>      <int>      <num>
#> 1:          1      0.067
#> 2:          2      0.100
#> 3:          3      0.067
```

